Year 12 Maths - Proof and mathematical communication

Similar questions - Proving inequalities and geometric results

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Prove that, for all real a and b, $a^2 + b^2 \geqslant 2ab$. **[2]**

(b) Hence prove that, for positive real numbers a and b, $\dfrac{a+b}{2} \geqslant \sqrt{ab}$. **[3]**

(c) Deduce that $a + \dfrac{1}{a} \geqslant 2$ for every $a > 0$, and state when equality occurs. **[3]**

2 $ABCD$ is any quadrilateral, with position vectors **a**, **b**, **c** and **d**. P, Q, R and S are the midpoints of AB, BC, CD and DA respectively.

(a) Find $\overrightarrow{PQ}$ and $\overrightarrow{SR}$ in terms of **a** and **c**. **[3]**

(b) Hence prove that $PQRS$ is a parallelogram. **[3]**

Question 1

(a) $a^2+b^2-2ab=(a-b)^2 \geqslant 0$, so $a^2+b^2 \geqslant 2ab$

(b) Apply (a) to $\sqrt{a}$ and $\sqrt{b}$ (both real as $a,b>0$): $a+b \geqslant 2\sqrt{ab}$. Divide by 2

(c) Take $b=\dfrac{1}{a}$: $\dfrac{a+\frac{1}{a}}{2} \geqslant \sqrt{a\times\frac{1}{a}}=1$, so $a+\dfrac{1}{a} \geqslant 2$. Equality when $a=\dfrac{1}{a}$, i.e. $a=1$

Question 2

(a) $\mathbf{p}=\frac{1}{2}(\mathbf{a}+\mathbf{b})$, $\mathbf{q}=\frac{1}{2}(\mathbf{b}+\mathbf{c})$, so $\overrightarrow{PQ}=\mathbf{q}-\mathbf{p}=\frac{1}{2}(\mathbf{c}-\mathbf{a})$. $\mathbf{r}=\frac{1}{2}(\mathbf{c}+\mathbf{d})$, $\mathbf{s}=\frac{1}{2}(\mathbf{d}+\mathbf{a})$, so $\overrightarrow{SR}=\mathbf{r}-\mathbf{s}=\frac{1}{2}(\mathbf{c}-\mathbf{a})$

(b) $\overrightarrow{PQ}=\overrightarrow{SR}$, so PQ and SR are equal in length and parallel; a quadrilateral with a pair of opposite sides equal and parallel is a parallelogram